ABGEN: Evaluating Large Language Models in Ablation Study Design and Evaluation for Scientific Research

Anonymous ACL submission

Abstract

We introduce **ABGEN**, the first benchmark designed to evaluate the capabilities of LLMs in designing ablation studies for scientific research. ABGEN consists of 2,000 expert-annotated examples derived from 677 NLP papers. In this benchmark, LLMs are tasked with generating detailed ablation study designs for a specified module or process based on the given research context. Our evaluation of leading LLMs, such as GPT-4o and Llama-3.1, highlights a significant performance gap between these models and human experts in terms of the importance, faithfulness, and soundness of the ablation study designs. Moreover, we demonstrate that current automated evaluation methods are not reliable for our task, as they show a significant discrepancy when compared to human assessment. To better investigate this, we develop **ABGEN-EVAL**, a meta-evaluation benchmark designed to assess the reliability of commonly used automated evaluation systems in measuring LLM performance on our task. We investigate various LLM-based evaluation methods on ABGEN-EVAL, providing insights for future research on developing more effective and reliable LLM-based evaluation systems for complex scientific tasks.

1 Introduction

In empirical scientific fields, designing experiments and selecting the appropriate experimental settings often present considerable challenges and requires significant domain expertise. Oftentimes, scientists learn about the flaws in their experimental design and missing ablations after going through a peer review process, which involves domain experts carefully evaluating a scientific work. The complexity of tasks in experimental science underscores the need for innovative approaches to support researchers in optimizing their workflows. Meanwhile, LLMs have demonstrated remarkable capabilities across a range of tasks integral to scientific processes, such as reviewing manuscripts (D'Arcy et al., 2024; Du et al., 2024), scientific writing (Altmäe et al., 2023; Xu et al., 2024), scientific code generation (Liu et al., 2023b; Yang et al., 2024b). This raises a compelling question: **Can LLMs be effectively leveraged to assist scientists in the process of experimental design?**

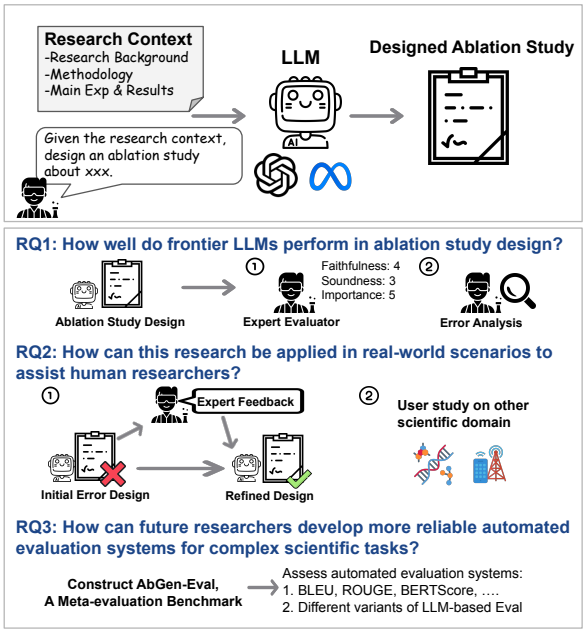

Figure 1: Overview of the research: the ablation study design task and three research questions investigated.

While addressing this question is inherently complex due to the diverse nature of scientific disciplines and difficulty of evaluation, our objective is to introduce the first comprehensive benchmark as well as an evaluation methodology to facilitate measuring progress on this task. We particularly introduce **ABGEN**[1], the first benchmark for evaluating LLMs in the context of designing ablation

[1]The ABGEN and ABGEN-EVAL benchmarks, codebase, and model outputs in this study, can be found at https://anonymous.4open.science/r/AbGen-7B40, and will be released publicly upon publication.

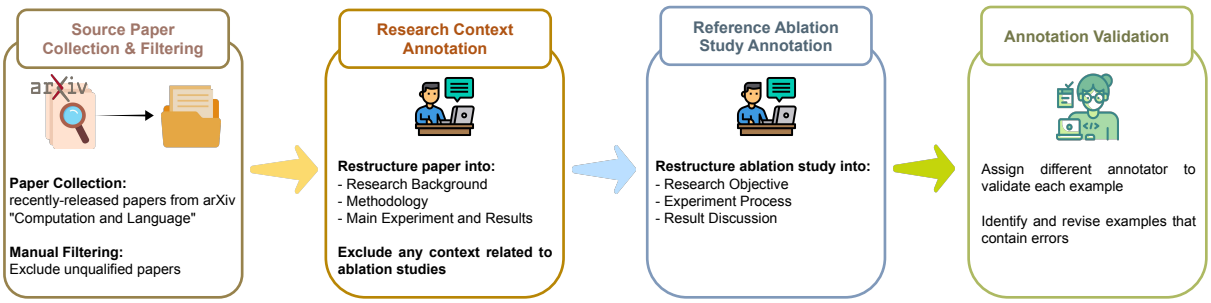

Figure 2: An overview of ABGEN construction pipeline.

studies for scientific research. The dataset consists of 2,000 examples derived from 677 scientific papers in natural language processing (NLP). Each example is carefully annotated and validated by NLP experts and includes a comprehensive research context along with a reference ablation study, both restructured from the original research paper. The research context is divided into three sections: research background, methodology, and the main experiment setup and results. As illustrated in Figure 1, the LLMs are tasked with generating a detailed ablation study design for a specified module or process based on the provided research context.

As outlined in Figure 1, we investigate three research questions in this study. Our main contributions are summarized below:

- We propose ABGEN, the first benchmark designed to evaluate the capabilities of LLMs in ablation study designs for scientific research (§2).
- We design a comprehensive human and automated evaluation systems for ABGEN (§3).
- We conduct a systematic evaluation of leading LLMs, analyzing their strengths and limitations on our new task, and providing insights for future advancements (§4.2).
- Our user studies reveals the potential of LLMs in ablation study design by interaction with human researchers, and highlights the adaptability of this approach to other scientific domains (§4.3).
- We develop the meta-evaluation benchmark, ABGEN-EVAL, and investigate various LLM-based evaluation methods to provide insights for creating more reliable automated evaluation systems for complex scientific tasks (§5).

2 ABGEN Benchmark

To systematically study the capabilities and limitations of current LLMs and measuring progress in assisting scientists with the design of their experimental workflows, we introduce a new benchmark named ABGEN. The LLMs are tasked with generating detailed ablation study designs for a specified module or process based on the given research context. We focus on scientific research within the NLP domain, as the involved expert annotators primarily have expertise in NLP (*i.e.,* each has at least one publication in a top-tier NLP or AI venue as a leading author). Detailed biographies of the annotators participating in the ABGEN annotation and LLM performance evaluation process are provided in Table 7 in Appendix A.

In the following subsections, we first provide a formal definition of the ABGEN task and then detail each step within the benchmark construction process. We present an overview of the ABGEN construction pipeline in Figure 2.

2.1 ABGEN Task Formulation

We formally define the task of ABGEN in the context of LLMs. Specifically, given:

- The **research context** $\boldsymbol{C}$, which is an expert-annotated context of a specific scientific study. This context is restructured from the original paper by expert annotators, including sections of research background, methodology, and main experiment setup and results (§2.3).
- The name of a specific essential module or process, denoted as $\boldsymbol{M}$, which is described in the *methodology* section within research context $\boldsymbol{C}$.

The LLM is tasked with generating the design for an ablation study, $\boldsymbol{A}$, aimed at evaluating the contribution and impact of $\boldsymbol{M}$ within the overall research framework:

$$\hat{\boldsymbol{A}} = \arg\max_{A} P_{\mathbf{LLM}}(\boldsymbol{A} \mid \boldsymbol{C}, \boldsymbol{M}) \qquad (1)$$

The ablation study design should include a clear

statement of the research objective, along with a detailed description of the experimental process.

2.2 Source Paper Collection and Filtering

Source Paper Collection We collect scientific papers from arXiv under the "Computation and Language" category, targeting those first released between March 1, 2024 and May 15, 2024—a timeframe likely after the cutoff date for data used in pretraining most current LLMs. This choice helps mitigate issues related to potential data memorization by the models being evaluated in this study. For each paper, we adopt the tool[2] developed by Lo et al. (2020) to extract its content. Specifically, this tool parses LaTeX source files of papers into JSON format, extracting features including the paper title, abstract, main sections, and appendix. We convert tables within the papers into HTML format. Both recent works (Sui et al., 2024; Fang et al., 2024) and our preliminary studies reveal that the evaluated LLMs can comprehend such table format effectively. We collect a total of 1,408 NLP papers as an initial pool for subsequent annotations. Next, we describe our approach and criteria for inclusion of the papers for annotation, as well as the details of the annotation process.

Research Paper Manual Filtering For each collected NLP paper, the expert annotator first determines if they are familiar with the paper's topic. If not, we randomly assign the paper to another annotator. Out of 1,408 papers, 217 were excluded as none of the expert annotators were familiar with their topics (*e.g.,* linguistic theories, spoken language processing, discourse and pragmatics, cognitive modeling), and 93 were excluded due to document parsing errors. The annotators are then instructed to determine whether the paper qualifies for inclusion in our benchmark. Specifically, we exclude: (1) Papers that are not focused on experimental work (e.g., surveys, position papers, dissertations), as they do not involve ablation study design; (2) Papers with fewer than two ablation studies, as these may not provide sufficient breadth of experimental evidence. Additionally, annotators may exclude papers they deem to be of low quality based on their expert judgment. After applying these filtering criteria, 677 papers remain for further annotation.

[2] https://github.com/allenai/s2orc-doc2json

2.3 Research Context Annotation

After determining that a research paper qualifies for benchmark inclusion, annotators are instructed to restructure the original paper into research context that maintains the original meaning but exclude any content related to ablation studies. The research context contains the following three sections: (1) **Research Background**, which is restructured from the introduction and related work sections, describing the paper's motivation, research problem, and relevant prior work. (2) **Methodology**, which is restructured from the methodology sections, This section describes the proposed method or model, including key components and innovations. (3) **Main Experiment Setup and Results**, which is restructured from the experiment sections. This section details the primary experimental setup, including datasets, baselines, and evaluation metrics used in main experiments, as well as the main experimental results. Appendix A and Figure 5 presents detailed examples of converting a research paper into research context.

2.4 Reference Ablation Study Annotation

Annotators are then tasked with restructuring each ablation study in the research paper into a reference ablation study. It consists of the following three sections: (1) **Research Objective**, a one- or two-sentence description of the research problem and the goal of the ablation study. If this statement is not explicitly provided in the original ablation study, annotators are required to infer and summarize it. (2) **Experiment Process**, a detailed account of the experimental setup, including the experimental groups, datasets, procedures, and the evaluation tools and metrics used. Annotators are requried to ensure that the process is clearly understandable and replicable based on the provided description. (3) **Result Discussion**, an analysis of the outcomes, where annotators summarize the key findings and their implications. It's worth noting that we do not instruct LLMs to generate this part in our experiments, as our main focus is on evaluating their ability to design ablation studies rather than execute and analyze experiments. However, we believe these features could be valuable for future research, which we elaborate on in the limitations section. Appendix A and Figure 8 present detailed examples of how an ablation study from a research paper is converted into a reference ablation study by expert annotators.

Property	Value
Research Context Word Length (avg./max)	1,481.1 / 2,207
Research Background (avg./max)	493.2 / 832
Methodology (avg./max)	513.5 / 1,023
Exp Setup & Results (avg./max)	477.4 / 921
Ref. Ablation Study Word Length (avg./max)	155.2 / 289
Research Objective (avg./max)	19.4 / 32
Experiment Process (avg./max)	72.3 / 172
Result Discussion (avg./max)	63.5 / 146
# NLP Research	677
# Ref. Ablation Study per Research (avg./max)	2.8 / 5
ABGEN Size	2,000
Testmini Set	500
Test Set	1,500

Table 1: Data statistics of the ABGEN benchmark.

2.5 Annotation Validation

For each annotated example, we assign an annotator to validate the annotated research context and reference ablation study based on the original research paper. They are required to identify and revise examples that contain errors. Out of the 2,000 annotated examples, 273 were identified as erroneous and were subsequently revised. We conducted a final human evaluation of data quality on 100 examples. As shown in Table 6 (Appendix A), for each validation metric, over 95% of the samples received a satisfaction rating of at least 4 out of 5. This result indicates the high quality of our dataset annotations.

2.6 Data Statistics

Table 1 illustrates the data statistics of the ABGEN benchmark. We randomly split the dataset into two subsets: *testmini* and *test*. The *testmini* subset contains 500 examples and is intended for both method validation and human analysis and evaluation. The *test* subset comprises the remaining 1,500 examples and is designed for standard evaluation. To prevent data contamination (Shi et al., 2024; Deng et al., 2024), only the ground truth for the *testmini* dataset will be publicly released. For the *test* subset, we will develop and maintain an online evaluation platform, allowing researchers to evaluate their methods and participate in a leaderboard.

3 ABGEN Evaluation

The automated evaluation of LLM generation for tasks relevant to scientific workflows remains an unsolved problem in the community. Recent benchmark work, such as SCIMON (Wang et al., 2024a) for novel scientific direction generation and MARG (D'Arcy et al., 2024) for peer review generation, primarily rely on human evaluation to assess LLM-based system performance. In our study, we also employ human evaluation by expert annotators as the *primary* assessment method. Additionally, in Section 5, we investigate different variants of LLM-based evaluation methods, aiming to provide insights for future work to develop automated evaluation systems for a large-scale evaluation.

3.1 Evaluation Criteria

This section discusses the human and automated evaluation protocols developed for ABGEN evaluation. We assess the following three dimensions for the generated ablation study design.

- **Importance**: The generated ablation study design will provide valuable insights into understanding the role of the specified module or process within the overall methodology.
- **Faithfulness**: The generated ablation study design aligns perfectly with the given research context. There are no contradictions between the generated content and the main experimental setup within the provided research context.
- **Soundness**: The generated ablation study design is logically self-consistent without ambiguious description. The human researchers would be able to clearly understand and replicate the ablation study based on the generated context.

To determine these three dimensions, we gathered feedback from three external senior NLP researchers, all of whom serve as area chairs for the ACL Rolling Review. Through iterative discussions, we identified these dimensions as critical for evaluating the quality and utility of generated ablation study designs. This feedback process also helped us in refining the assessment guidelines used for human evaluation (§3.2). We do not evaluate the *fluency* of the generated ablation study, as both recent works (D'Arcy et al., 2024; Zeng et al., 2024) and our preliminary findings find that leading LLMs consistently produce fluent text free of grammatical errors.

3.2 Human Evaluation Protocol

In this subsection, we first outline the process of human evaluation, followed by a discussion of our methods to ensure the reliability and reproducibility of the evaluation protocol.

Human Evaluation Process For human evaluation, we use Likert-scale scores ranging from 1 to 5 for each criterion (*i.e.,* importance, faithfulness, and soundness). Given the research context and an LLM-generated ablation study, human evaluators are asked to score the generated content for each criteria. Initially, the reference ablation study is not provided to the evaluator. This approach encourages evaluators to carefully review the generated content in light of the research context, reducing the likelihood of bias from comparing it to the reference. This is particularly important, as LLMs may generate ablation studies that, while reasonable, differ from the reference. For any criterion where the evaluator does not assign a score of 5, they are required to provide a one- or two-sentence justification for their decision. After submitting their initial scores, evaluators are then given the reference ablation study and asked to adjust their scores if they identify any aspects they may have initially overlooked.

Ensuring Reliable and Reproducible Human Evaluation To ensure the reliability and reproducibility of our human evaluation protocol, we develop a comprehensive human assessment guideline.[3] In the guideline, for each evaluation criterion, we provide detailed definitions accompanied by one or more examples to clearly explain the basis for each score. Expert annotators are required to thoroughly review the guideline document and successfully complete a pilot annotation task before participating in the formal evaluation. To assess inter-annotator agreement of our human evaluation, we sample 40 fixed LLM-generated outputs that are separately evaluated by all four expert annotators. They achieve inter-annotator agreement scores (*i.e.,* Cohen's Kappa) of 0.735, 0.782, and 0.710 for the criteria of importance, faithfulness, and soundness, respectively. These results demonstrate a high level of consistency among the different evaluators involved in human evaluation.

3.3 Automated Evaluation

While human evaluation is considered reliable, it is time-consuming and not easily scalable. Therefore, we systematically investigate several automated evaluation systems to study their efficacy for this task. We first adopt traditional evaluation metrics, including **ROUGE** (Lin and Hovy, 2003), **BLEU** (Papineni et al., 2002; Post, 2018), and **BERTScore** (Zhang* et al., 2020) for evaluation. We also explore the use of LLM-based evaluation systems. Specifically, we investigate following two popular approaches: (1) **LLM Pointwise Scoring** (Liu et al., 2023a; Chiang and Lee, 2023), which evaluates the quality of the generated ablation study by assigning an integer score between 1 and 5. (2) **LLM Pairwise Comparison** (Zheng et al., 2023; Wang et al., 2023), which evaluates two outputs generated by different LLMs to determine which is superior or declare a tie. We developed different variants for both LLM-based evaluation approaches to explore the impact of factors such as providing reference ablation studies, offering fine-grained evaluation instructions, and using Chain-of-Thought (CoT) reasoning, as detailed in Section 5. We select the most effective variant to report scores in Table 2, with its prompt presented in Figure 18 in Appendix C.

4 LLMs for Ablation Study Design

4.1 Experiment Setup

Evaluated Systems We examine the performance of 9 frontier LLMs across two distinct categories on our benchmark: (1) **Proprietary LLMs**, including GPT-4o and GPT-4o-mini (OpenAI, 2024), Gemini-1.5-Pro (Gemini, 2024), and Claude-3.5-Sonnet (Anthropic, 2024); and **Open-source LLMs**, including Llama-3.1-70 & 405B[4] (AI@Meta, 2024), Mistral-Large (Jiang et al., 2024), Deepseek-V2 (DeepSeek-AI, 2024), and Qwen2-72B (Yang et al., 2024a). Table 8 in Appendix B presents the details of these evaluated LLMs in ABGEN. We also evaluate the performance of reference ablation study design and human experts, with the process detailed in Appendix A.1. The results are presented in Table 2.

Implementation Details For all the experiments, we set temperature as 1.0 and maximum output length as 1024 (as the maximum length of reference ablation study is 289 words). Figure 3 in Appendix B illustrates the default prompt used across all generation experiments. The model is tasked with generating the design for an ablation study, based on the provided annotated research context and the specified module or process name. Specifically, the LLMs are required to first generate a

[3]The human assessment guideline for the ABGEN human evaluation protocol is available at this anonymous link.

[4]Due to computational constraints, for the Llama-3.1-405B model, we use the API service from Google Vertex AI.

System	Traditional Metrics			LLM-based Eval (1-5)				Human Evaluation (1-5)			
	BLEU	R-L	BertS	Import.	Faith.	Sound.	Overall	Import.	Faith.	Sound.	**Avg.**
Reference Ablation Study	–	–	–	–	–	–	–	4.70	4.90	4.70	4.77
Human Expert	14.9	27.4	88.7	4.12	4.15	4.43	4.37	4.65	4.93	4.83	4.80
GPT-4o	9.44	22.4	73.3	3.89	3.38	3.31	3.37	3.87	3.40	3.28	3.34
Claude-3.5-Sonnet	8.51	19.4	71.2	3.66	3.20	3.14	3.15	3.83	3.20	3.31	3.26
Gemini-1.5-pro	10.52	23.6	84.8	4.02	3.46	3.63	3.37	3.69	3.10	3.33	3.21
Llama-3.1-405B	8.89	22.6	74.7	3.55	3.02	3.15	3.14	3.60	2.99	3.04	3.02
GPT-4o-mini	9.56	22.1	77.0	3.84	3.37	3.74	3.45	3.62	2.88	3.11	3.00
Mistral-Large	8.92	20.4	72.1	3.71	3.23	3.26	3.12	3.46	2.84	2.97	2.91
Deepseek-V2	9.72	24.2	81.0	3.78	3.26	3.68	3.18	3.11	2.97	2.81	2.89
Llama-3.1-70b	8.71	19.5	72.3	3.40	2.86	2.88	3.24	3.26	2.75	2.75	2.75
Qwen2-72b	8.96	20.8	70.9	3.25	2.92	2.80	3.09	3.04	2.63	2.56	2.59

Table 2: Human and automated evaluation results of the frontier LLMs on ABGEN *testmini* set. For the LLM-based evaluation scores, we use the best-performing GPT-4o-mini variant based on performance on our ABGEN-EVAL meta-evaluation benchmark (§5.2). For human evaluation, we randomly sample 100 examples from the *testmini* subset. For each model output, an expert evaluator is assigned to conduct the human evaluation. The average scores of the human evaluations is used as the ranking indicator of model performance for this table.

one-sentence description of the research objectives, followed by a detailed description of the experimental setup for the ablation study.

4.2 Results and Analysis

> 💡 **RQ1:** How well do frontier LLMs perform in designing ablation studies?

Table 2 illustrates the performance of the evaluated LLMs on the ABGEN *testmini* set, while Table 9 in Appendix B shows their performance on the *test* set. The human evaluation results demonstrate that ABGEN poses significant challenges to current LLMs. Even the best-performing LLM, GPT-4o, performs much worse than human experts. This gap highlights the critical need for further advancements in LLMs, especially in applying them to complex scientific tasks. Open-source LLMs still lag behind the frontier proprietary LLMs. Even LLama-3.1-405B cannot match the performance of proprietary models. Moreover, we observe a significant disparity between automated evaluation systems and human assessments. For instance, despite receiving lower scores in LLM-based evaluations compared to Claude-3.5-Sonnet, GPT-4o consistently outperforms it in every criterion according to human evaluation. Additionally, traditional reference-based metrics fail to provide reliable rankings for the evaluated LLMs. These results suggest that current automated evaluation systems may not be fully reliable for our task.

Error Analysis We further conduct a comprehensive error analysis to better understand the capabilities and limitations of the top-performing LLMs on our task. This error analysis is based on 100 failure cases of GPT-4o from the *testmini* set, where the average human evaluation scores are below 3. We identify five common error types, and provide detailed explanations for each type in Table 3. See Appendix B for examples of each error type.

4.3 User Studies on Real-world Scenarios

> 💡 **RQ2:** How can this research be applied in real-world scenarios to assist human researchers in designing ablation studies?

To investigate this research question, we design and conduct following two user studies:

LLM-Researcher Interaction While LLMs currently lag behind human experts in designing ablation studies, they still hold value as tools to assist researchers. To explore this potential, we examine scenarios where researchers interact with LLMs, providing feedback to guide the refinement of their outputs. Specifically, we first sample 20 failure cases from *testmini* set—each with an average human score below 3—from both GPT-4o and Llama-3.1-70B. Two expert annotators are then tasked with reviewing these LLM-generated ablation study designs, identifying errors, and providing constructive feedback for improvement within a 50-word limit. We then feed the research context, initial ablation study design, and researcher feedback back into the same LLMs, instructing them to regenerate the ablation study design. Another expert evaluator is then assigned to assess the revised

Error Type	Explanation
Misalignment with research context (36/100)	This error arises when the generated experiment process contradicts with the baseline in the research context or introduces factual errors. (**Example:** Figure 11)
Ambiguity and Difficulty in Reproduction (34/100)	This error arises when the generated experiment process contains ambiguous steps or lacks the necessary datasets or tools, for human researchers to replicate the ablation study. (**Example:** Figure 12)
Partial Ablation or Incomplete Experimentation (28/100)	This error arises when the generated experiment process partially addresses the ablation module, such as only ablating a sub-module, or missing experimental groups. (**Example:** Figure 13)
Insignificant Ablation Module (27/100)	This error arises when the generated research objective is focused on an insignificant ablation module in research context. (**Example:** Figure 14)
Inherent Logical Inconsistencies (22/100)	This error arises when the generated experiment process contains inherent logical inconsistencies, such as gaps in implementation steps. (**Example:** Figure 15)

Table 3: Error analysis of GPT-4o's failure cases. We provide examples of each error type in Appendix B.

User Study	Import.	Faith.	Sound.
User Study 1: LLM-Researcher Interaction			
GPT-4o			
Initial Failure Case	3.9	2.1	2.0
Revision with Feedback	4.8 (+0.9)	4.2 (+2.1)	4.6 (+2.6)
Llama-3.1-70B			
Initial Failure Case	3.7	1.8	1.7
Revision with Feedback	4.5 (+0.8)	3.9 (+2.1)	4.1 (+2.4)
User Study 2: Domain Generalization			
GPT-4o			
NLP Domain (as Main Exp)	3.9	3.4	3.3
Biomedical Domain	3.7	3.4	3.1
Computer Network Domain	3.8	3.3	3.4
Llama-3.1-70B			
NLP Domain (as Main Exp)	3.3	2.8	2.8
Biomedical Domain	3.0	2.8	2.9
Computer Network Domain	3.1	2.9	3.0

Table 4: Human evaluation result from two user studies. The findings demonstrate (1) the potential of LLMs in designing ablation studies through interaction with human researchers, and (2) the adaptability of our research across different scientific domains.

version, following the same human evaluation protocol in Section 3.2. We present the prompts used, along with examples of the initial generation, the corresponding research feedback, and the refined ablation study design in Figure 16 in Appendix B. As shown in Table 4, incorporating researcher feedback can significantly enhance LLM performance in refining their outputs.

Domain Generalization of Our Research Our research primarily focuses on NLP domains. To explore the adaptability of our work across other scientific fields, we conducted user studies in the areas of biomedical sciences and computer networks. Specifically, we engage two experts—one in computer networking and one in biomedical research—to provide five research papers from their respective fields that were first published after May 1, 2024, and with which they are familiar. Following the same procedure as ABGEN annotation, they annotate the research context and reference ablation studies from five corresponding papers, resulting in a total of 27 examples over ten papers. We then provide them with LLM-generated ablation study designs and ask them to strictly follow our human assessment guidelines to evaluate the LLM outputs. As shown in Table 4, the human evaluation scores for GPT-4o and Llama-3.1-70B are consistent with the results observed in the NLP domain experiments. We believe that future work could extend our research framework to other scientific domains.

5 Investigating Automated Evaluation for Ablation Study Design

RQ3: How can future researchers develop more reliable and effective automated evaluation systems for complex scientific tasks?

As discussed in Section 4.2, we observe a significant discrepancy between automated and human evaluation results when assessing LLM performance on ABGEN. To investigate this issue further, we conduct a systematic meta-evaluation of commonly used automated evaluation systems.

5.1 ABGEN-EVAL Benchmark

We construct the meta-evaluation benchmark, ABGEN-EVAL, based on the human assessments results collected in Section 4. ABGEN-EVAL comprises 9 LLM outputs $\times$ 100 human assessments

= 900 examples. Each example includes: (1) an LLM-generated ablation study design; (2) three human scores assessing the study's importance, faithfulness, and soundness, respectively; and (3) a one- or two-sentence justification for each criterion with a score below 5 (detailed in §3.2). In line with previous meta-evaluation studies (Fabbri et al., 2021; Chen et al., 2021; Liu et al., 2024), in ABGEN-EVAL, the human evaluation results on the system-generated ablation study is considered the gold standard. The performance of automated evaluation systems is measured by the **system-level** and **instance-level** correlation (with detailed definitions in Appendix C) between scores of human evaluation and automated evaluation systems.

5.2 Experiments

For the two LLM-based evaluation systems (*i.e.,* LLM Pointwise Scoring and LLM Pairwise Comparison), we developed different variants to explore how various factors impact their effectiveness. These factors include (1) providing reference ablation studies, (2) offering fine-grained evaluation instructions, (3) using CoT reasoning, (4) using different LLMs (*e.g.,* GPT-4o and GPT-4o-mini) as the base evaluators, and (5) evaluate using a specific creteria or overall scores. We present the prompt for each variant shown from Figure 18 to Figure 23 in the Appendix. Table 5 shows the system-level correlation between the automated evaluation systems under investigation and human assessments, while Table 10 in the Appendix presents the instance-level results. The current automated evaluation systems show relatively low correlations, especially at the instance level, indicating that they are not reliable for assessing generated ablation study designs. Among LLM-based evaluation variants, the reference-based approach consistently outperforms the reference-free approach. Additionally, using CoT prompting and fine-grained instructions can further improve the effectiveness of LLM-based evaluations. Furthermore, the variant using GPT-4o demonstrates higher correlations with human evaluations at both the system and instance levels compared to the GPT-4o-mini variant. This suggests that employing more advanced LLMs as the base evaluator can enhance its reliability.

Automated Evaluation Systems	Faith.	Sound.	Import.	Avg.
Traditional Metrics				
BLEU	0.256	0.329	0.107	0.231
ROUGE-L	0.295	0.188	0.004	0.162
BERTScore	0.203	0.304	0.043	0.183
LLM Pointwise Scoring				
GPT-4o-mini (Criteria-specific)				
Ref. + Fine-grained instruct + CoT	0.645	0.490	0.592	0.576
Fine-grained instruct + CoT	0.399	0.600	0.468	0.489
Fine-grained instruct	0.540	0.387	0.438	0.455
Ref. + Fine-grained instruct	0.157	0.574	0.594	0.442
Ref. + Coarse instruct	0.403	0.475	0.445	0.441
Ref. + Coarse instruct + CoT	0.329	0.495	0.448	0.424
Coarse instruct + CoT	0.287	0.413	0.573	0.424
Coarse instruct	0.398	0.364	0.503	0.422
GPT-4o-mini (Overall Scores)				
Ref. + Fine-grained instruct + CoT	0.396	0.540	0.491	0.476
GPT-4o (Criteria-specific)				
Ref. + Fine-grained instruct + CoT	0.724	0.496	0.756	0.659
Fine-grained instruct + CoT	0.546	0.576	0.713	0.612
Fine-grained instruct	0.577	0.570	0.685	0.611
LLM Pairwise Comparison				
GPT-4o-mini (Criteria-specific)				
Ref. + Fine-grained instruct + CoT	0.471	0.607	0.445	0.508

Table 5: Pearson correlations at the system level between automated evaluation and human assessments. The GPT-4o-mini "Ref. + Fine-grained instruct + CoT" is used for reporting scores in Table 2.

6 Related Work

LLMs have been employed for different scientific tasks for enhancing researchers' scientific workflows, such as conducting literature reviews (Wang et al., 2024b; Agarwal et al., 2024), peer-review and meta-review generation (D'Arcy et al., 2024; Tan et al., 2024; Wu et al., 2022; Zhou et al., 2024a), question answering over scientific papers (Dasigi et al., 2021; Saikh et al., 2022; Lee et al., 2023), scientific paper writing (Xu et al., 2024) and research hypothesis generation (Wang et al., 2024a; Zhou et al., 2024b). However, the potential of LLMs to effectively assist scientists in the experimental design process remains largely unexplored. Additionally, the challenge of developing effective and reliable automated evaluation systems for complex scientific tasks is still an open question.

7 Conclusion

This paper introduces ABGEN, the first benchmark designed to evaluate LLMs in generating ablation studies for scientific research. Through a comprehensive assessment, we highlight both the strengths and limitations of leading LLMs on ABGEN, providing valuable insights for future advancements. We also develop a meta-evaluation benchmark, providing insights into developing more reliable automated evaluation for complex scientific tasks.

Limitations

In this work, we introduce ABGEN and perform a comprehensive analysis of various LLMs' capabilities on the proposed new task. However, there are still some limitations:

First, this study does not explore advanced prompting techniques (Yao et al., 2023; Wang et al., 2024a) or LLM-Agent-based methods (D'Arcy et al., 2024; Majumder et al., 2024). Our focus is on assessing the fundamental capabilities of leading LLMs in ablation study design. The goal is to provide insights into their strengths and limitations, laying the groundwork for future advancements. We encourage researchers to build upon our benchmark and findings to develop more advanced approaches for this task.

Second, as shown in our results on ABGEN-EVAL, the reported automated evaluation scores are not yet perfect. To support further research, we will make all model outputs from Section 4 publicly available. This will enable other researchers to conduct different automated evaluations and ensure consistent rankings by re-running their assessments on our model outputs. Additionally, our human evaluation protocol is designed to minimize the need for repeated human evaluations by future researchers. By strictly adhering to our assessment guidelines, researchers can reliably assess and compare their methods with existing approaches in an independent and consistent manner.

Lastly, we do not evaluate LLMs specialized in scientific domains (Groeneveld et al., 2024; Wadden et al., 2024), as our preliminary experiments find that they struggle to follow the instructions to generate ablation studies. However, we believe that training LLMs on scientific data (Lo et al., 2020) could enhance their performance on our task.

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A ABGEN Benchmark

Annotation Quality	**%S $\geq$ 4**
Research Context	
Correctly structured	99.0
Excluding ablation-relevant content	96.5
Reference Ablation Study	
Correctly structured	98.5
Non-overlapping	96.0
Justifiable within research context	97.5

Table 6: Human evaluation over 200 samples of ABGEN. Three internal evaluators were asked to rate the samples on a scale of 1 to 5 individually. We report percent of samples that have an average score $\geq$ 4 to indicate the annotation quality of ABGEN.

A.1 Performance of Real Paper and Expert

To provide an informative estimate of real paper and expert-level performance on ABGEN, we randomly sample 20 examples from 10 papers in the *testmini* set. We enlist two expert annotators (*i.e.,* Annotators 1 and 4, as described in Table 7 in Appendix A) to individually solve these examples. To ensure fairness, we mix these 20×2 expert-annotated data and corresponding 20 reference ablation study within the standard human evaluation process. The expert evaluators are not informed of the sources of these ablation study examples when evaluation. We report the evaluation results on Table 2.

B Experiments

B.1 Experiment Setup

ID	# NLP/AI Publication	Data Annotation	Data Validation	Human Evaluation	Human Performance
1	> 10	✓	✓		✓
2	> 10			✓	
3	> 10			✓	
4	5-10	✓	✓		✓
5	1-5	✓		✓	
6	1-5	✓	✓	✓	

Table 7: Details of annotators involved in dataset construction and LLM performance evaluation. ABGEN is annotated by experts in NLP domains, ensuring both the accuracy of the benchmark and the reliability of the human evaluation.

Model	Organization	Release	Max Len.	Source & Model Version
GPT-4o (OpenAI, 2024)	OpenAI	2024-05	128K	`gpt-4o-2024-05-13` (API Documentation)
GPT-4o-mini (OpenAI, 2024)	OpenAI	2024-07	128K	`gpt-4o-mini-2024-07-18` (API Documentation)
Gemini-1.5-Pro (Gemini, 2024)	Google	2024-02	2M	API Documentation
Claude-3.5-Sonnet (Anthropic, 2024)	Anthropic	2024-06	200k	https://www.anthropic.com/api `claude-3-5-sonnet-20240620`
Llama-3.1-405B (AI@Meta, 2024)	Meta	2024-07	128K	Google Vertex AI (API Documentation) `llama3-405b-instruct-maas`
Llama-3.1-70B (AI@Meta, 2024)	Meta	2024-07	128K	`meta-llama/Meta-Llama-3.1-70B-Instruct`
Qwen-2-72B (Yang et al., 2024a)	Qwen	2024-06	128K	`Qwen/Qwen2-72B-Instruct`
Mistral-Large (Jiang et al., 2024)	Mistral AI	2024-07	128K	`mistralai/Mistral-Large-Instruct-2407`
Deepseek-V2 (DeepSeek-AI, 2024)	DeepSeek	2024-05	128K	`deepseek-ai/DeepSeek-V2-Chat`

Table 8: Details of the organization, release time, maximum context length, and model source (*i.e.,* url for proprietary models and Huggingface model name for open-source models) for the LLMs evaluated in ABGEN.

Ablation Generation Prompt

`[System Input]:`
Given the research context, design an ablation study for the specified module or process. Begin the design with a clear statement of the research objective, followed by a detailed description of the experimental process.

`[User Input]:`
Research Context:{research context}
Design an ablation study about {ablation module} based on Research Context:

Figure 3: Prompt for ablation study generation.

User Study Prompt

`[System Input]:`
Revise or rewrite the initial generation based on research context and user feedback.

`[User Input]:`
Research context: {research context}
Initial generation: {initial generation}
User feedback: {user feedback}

Redesign an ablation study about the {ablation module}, according to user feedback . . .

Figure 4: Prompt for user study about LLM-researcher interaction.

Model	Traditional Metrics			LLM-based Eval (1-5)			
	BLEU	R-L	BertS	Import.	Faith.	Sound.	Overall
GPT-4o	9.86	22.3	74.5	3.86	3.43	3.34	3.38
Claude-3.5-Sonnet	8.55	19.7	70.1	3.69	3.24	3.12	3.17
Gemini-1.5-pro	10.68	23.3	82.1	4.05	3.45	3.65	3.33
Llama-3.1-405B	8.59	21.9	75.8	3.56	3.05	3.18	3.18
GPT-4o-mini	9.53	22.4	77.3	3.88	3.34	3.72	3.45
Mistral-Large	8.68	20.2	73.6	3.69	3.26	3.28	3.11
Deepseek-V2	9.83	25.6	82.3	3.72	3.30	3.69	3.22
Llama-3.1-70b	8.64	21.1	73.2	3.28	2.88	2.92	3.31
Qwen2-72b	8.90	20.9	69.7	3.22	2.97	2.80	3.06

Table 9: Automated evaluation results of the frontier LLMs on ABGEN *test* set. For the LLM-based evaluation scores, we use the best-performing variant based on performance on our ABGEN-EVAL meta-evaluation benchmark (§5.2). The overall scores of the LLM-based Eval is used as the ranking indicator of model performance for this table.

Research Context Annotation — Research Background

Paper Title: Source-Aware Training Enables Knowledge Attribution in Language Models

url: https://arxiv.org/pdf/2404.01019v2

Introduction: Large language models (LLMs) often generate content that is not based on factual information. As LLMs are pretrained over noisy web data that often contains inaccurate or outdated content, users should be able to verify LLM outputs by checking their sources. Moreover, concerns about copyright infringement, privacy violations, data contamination, and toxic content in LLMs emphasize the need for techniques to identify and trace the origins of information included in models' responses. It is therefore desirable if LLMs can provide supporting evidence for their responses by citing or attributing the outputs to the sources they draw upon. Beyond improving the models' transparency, attribution allows for a deeper understanding of the relationship between training data and model behaviors, thereby offering a pathway to refine the quality of pretraining data. We focus on intrinsic source citation, where the LLM should cite source documents from the pretraining data from which it acquired its relevant parametric knowledge. Compared to retrieval-based approaches such as RAG or post-hoc techniques, intrinsic source citation is inherently tied to the model itself, enables more faithful attribution to its parametric knowledge, thus opens up unique opportunities for improved interpretability. To this end, we explore source-aware training—a post-pretraining recipe that enables a LLM to cite its pretraining data based on its parametric knowledge. Our motivation is three-fold. First, a significant portion of an LLM's knowledge is acquired during pretraining, therefore citing evidence for this parametric knowledge can greatly enhance the LLM trustworthiness. Second, the standard practice for LLM pretraining neglects the attribution angle, which explains why the current generation of LLMs fails to provide reliable citations. We aim to explore a training procedure that naturally facilitates citation of the pretraining data. Finally, from a scientific perspective, it is intriguing to investigate whether and how current language models can be trained to reference their pretraining data. We inquire: Given an off-the-shelf LLM, can we train it to attribute its generations to the supporting sources from the pretraining data? Our goal is to cite the pretraining documents themselves. (Our setup mirrors existing frameworks for LLM pretraining and can be summarized as follows: …, **proposed method and experiment process, removed.**) (Our contributions are summarized as follows: … **the contribution of this paper, removed.**)

Related Work: Language Model Attribution. Attribution is gaining more attention recently as interpretability and grounding of language models become increasingly important. Generally speaking, approaches to achieve attribution can be classified as either retrieval-based or model-based. Retrieval-based approaches include retrieval augmentation (RAG) and post-hoc attribution. RAG approaches enable attribution by providing a retrieved context for the LM to use, and teaching LM how to cite the retrieved context. The major limitations of RAG approaches are the lack of guarantee that the model is relying on the retrieved data for generation, and that they only work on non-parametric knowledge. Post-hoc approaches attribute the LM outputs by retrieving the supporting evidence given the model's response, but have been shown to produce non-accurate citations. Model-based techniques involve prompting the model directly to generate citations for its parametric knowledge or scaling techniques such as influence functions to large models. Modelbased attribution is arguably more faithful than retrieval-based approaches as the citation mechanism is intrinsic to the model. However, standard approaches to pretraining LMs do not take into account the need for the language model to cite its pretraining data (, which is where our work comes into play, **the contribution of this paper, removed**). Bohnet et al. proposed the task of attributed question-answering and evaluated the attribution performance of different systems using the AutoAIS metric. In addition, they fine-tuned PaLM to generate both an answer and a URL pointing to Wikipedia page supporting the answer in generative retrieval style. (Although this setup is similar to ours in that we require the LM to generate the document identifier as well, **proposed method and experiment process, removed**) their setup is basically a variation of RAG where the LM acts as the retriever. Citation Generation. There is a large body of work on the task of citation generation in the scientific domain, where the goal is to cite an appropriate article given a particular context or to generate text citing one article in relation to another. A relevant work to ours is Galactica, which leverages the underlying citation graph in the pretraining data to learn to predict citations given a context. Notably, Galactica is trained to leverage citations of scientific articles in the pretraining data (, while our work explores citation of all the pretraining documents, extending beyond scientific articles, **the contribution of this paper, removed**). Gao et al. introduced a benchmark for the automatic evaluation of LM citations and Ye et al. proposed a method to improve language model grounding by fine-tuning the language model on responses that are well supported by their citations. However, their setup is restricted to citation of retrieved rather than parametric knowledge. Generative Retrieval. Our work is somewhat related to generative retrieval, where an auto regressive model is trained to act as a retriever in an information retrieval (IR) system. Generative retrieval typically relies on a transformer model to map a given query to a document identifier that is likely to contain an answer to the query. (While our task also requires the language model to generate a document identifier, we differ from generative retrieval in at least two ways. First, … , **proposed method and experiment process, removed**)

Annotated Research Background:

Large language models (LLMs) often generate content that is not based on factual information. As LLMs are pretrained over noisy web data that often contains inaccurate or outdated content, users should be able to verify LLM outputs by checking their sources. Moreover, concerns about copyright infringement, privacy violations, data contamination, and toxic content in LLMs emphasize the need for techniques to identify and trace the origins of information included in models' responses. It is therefore desirable if LLMs can provide supporting evidence for their responses by citing or attributing the outputs to the sources they draw upon. Beyond improving the models' transparency, attribution allows for a deeper understanding of the relationship between training data and model behaviors, thereby offering a pathway to refine the quality of pretraining data. We focus on intrinsic source citation, where the LLM should cite source documents from the pretraining data from which it acquired its relevant parametric knowledge. Compared to retrieval-based approaches such as RAG or post-hoc techniques, intrinsic source citation is inherently tied to the model itself, enables more faithful attribution to its parametric knowledge, thus opens up unique opportunities for improved interpretability. To this end, we explore source-aware training—a post-pretraining recipe that enables a LLM to cite its pretraining data based on its parametric knowledge. Our motivation is three-fold. First, a significant portion of an LLM's knowledge is acquired during pretraining, therefore citing evidence for this parametric knowledge can greatly enhance the LLM trustworthiness. Second, the standard practice for LLM pretraining neglects the attribution angle, which explains why the current generation of LLMs fails to provide reliable citations. We aim to explore a training procedure that naturally facilitates citation of the pretraining data. Finally, from a scientific perspective, it is intriguing to investigate whether and how current language models can be trained to reference their pretraining data. We inquire: Given an off-the-shelf LLM, can we train it to attribute its generations to the supporting sources from the pretraining data? Our goal is to cite the pretraining documents themselves.

Language Model Attribution. Attribution is gaining more attention recently as interpretability and grounding of language models become increasingly important. Generally speaking, approaches to achieve attribution can be classified as either retrieval-based or model-based. Retrieval-based approaches include retrieval augmentation (RAG) and post-hoc attribution. RAG approaches enable attribution by providing a retrieved context for the LM to use, and teaching LM how to cite the retrieved context. The major limitations of RAG approaches are the lack of guarantee that the model is relying on the retrieved data for generation, and that they only work on non-parametric knowledge. Post-hoc approaches attribute the LM outputs by retrieving the supporting evidence given the model's response, but have been shown to produce non-accurate citations. Model-based techniques involve prompting the model directly to generate citations for its parametric knowledge or scaling techniques such as influence functions to large models. Modelbased attribution is arguably more faithful than retrieval-based approaches as the citation mechanism is intrinsic to the model. However, standard approaches to pretraining LMs do not take into account the need for the language model to cite its pretraining data. Bohnet et al. proposed the task of attributed question-answering and evaluated the attribution performance of different systems using the AutoAIS metric. In addition, they fine-tuned PaLM to generate both an answer and a URL pointing to Wikipedia page supporting the answer in generative retrieval style. Their setup is basically a variation of RAG where the LM acts as the retriever. There is a large body of work on the task of citation generation in the scientific domain, where the goal is to cite an appropriate article given a particular context or to generate text citing one article in relation to another. A relevant work is Galactica, which leverages the underlying citation graph in the pretraining data to learn to predict citations given a context. Notably, Galactica is trained to leverage citations of scientific articles in the pretraining data. Gao et al. introduced a benchmark for the automatic evaluation of LM citations and Ye et al. proposed a method to improve language model grounding by fine-tuning the language model on responses that are well supported by their citations. However, their setup is restricted to citation of retrieved rather than parametric knowledge. Generative Retrieval. Generative retrieval, where an auto regressive model is trained to act as a retriever in an information retrieval (IR) system, typically relies on a transformer model to map a given query to a document identifier that is likely to contain an answer to the query.

Figure 5: An example of **research context construction**

Research Context Annotation — Methodology
Paper Title: Source-Aware Training Enables Knowledge Attribution in Language Models
url: https://arxiv.org/pdf/2404.01019v2

Methodology: Source-Aware Training Our training framework is designed to easily integrate with existing pretraining pipelines. (We minimize its deviations from established post-pretraining practice, … **experiment setup description, removed**) Each document in the pretraining corpus is assigned a unique document identifier (ID) and our goal is to train a language model that can respond to user prompts by providing both a response and an ID referring to the source document of the model's knowledge. Setup. Our evaluation follows the attributed question answering setup, where given an input prompt z, the LLM output will consist of a tuple ⟨r, c⟩ where r is the response (e.g., the answer to a question) and c is the identifier of the document in the pretraining data that supports the answer.
Following standard LLM training setups, our recipe has two stages: Continual pretraining (Section 3.1) and instruction tuning (Section 3.2). Instruction tuning trains the model to be able to attribute the generated responses to supporting documents it has seen during pretraining. The pretraining stage will involve all documents by nature, but the instruction tuning step is restricted to a subset of the pretraining documents. This restriction is due to the potential cost of curating instruction tuning data from all the pretraining documents, that is in addition to the training overhead incurred by instruction tuning. After training, we measure out-of-domain (OOD) attribution: whether the model can attribute knowledge to documents that are only included in the continual pretraining data but not in the instruction tuning data. We therefore split the pretraining corpus into in-domain and OOD subsets. The in-domain data is used to create attribution training examples, while the OOD documents are used for evaluation.
Continual Pretraining with Doc ID Injection Doc ID injection. The continual pretraining phase has two goals: (i) memorizing knowledge via next-word prediction (same as established LLM pretraining), and (ii) associating knowledge within a source document with its ID to enable OOD attribution. We aim to achieve the second goal by injecting the document ID into the document before training. An important consideration is the location and frequency of injecting the document ID. Formally, given a pretraining corpus of documents {x (1) , x (2) , ..., x (m)} and their corresponding IDs {c (1) , c (2) , ..., c (m)} where each x (i) is a sequence of tokens x (i) = (x (i) 1 , x (i) 2 , ..., x (i) |xi |), and each c (i) = (c (i) 1 , c (i) 2 , ..., c (i) |ci |) is a sequence of tokens of its identifier. Our pretraining aims to learn the language model parameters θ that maximize the objective maxθ ∑ m i=1 log P(xˆ (i) ; θ) , where xˆ (i) is the ID-injected version of the document x (i) . We inject the doc ID into document a x with different strategies, each of which corresponds to a different xˆ. 2 (Particularly, we experiment with the following strategies: 1. NO-ID: Standard pretraining without ID injection: xˆ := x. 2. …, **experiment setup description, removed**)Attention masking to improve training efficiency. To maximize GPU utilization during continual pretraining, the typical practice packs several pretraining documents within a single training sequence separated by the end-of-sentence token. As a result, the doc ID tokens for a certain document will naturally attend to preceding tokens from other documents. (Our initial experiments showed that … **ablation study, removed**) (Instruction Tuning In addition to pretraining, we further adapt the model to … **ablation study, removed**) The setup in the previous section train the model to recall the doc ID immediately after the answer. (Another setup we explore is …, **ablation study, removed**)

Annotated Methodology:
Our source-aware training framework is designed to easily integrate with existing pretraining pipelines. Each document in the pretraining corpus is assigned a unique document identifier (ID) and our goal is to train a language model that can respond to user prompts by providing both a response and an ID referring to the source document of the model's knowledge. Our evaluation follows the attributed question answering setup, where given an input prompt z, the LLM output will consist of a tuple ⟨r, c⟩ where r is the response and c is the identifier of the document in the pretraining data that supports the answer.
Following standard LLM training setups, our recipe has two stages: Continual pretraining and instruction tuning. Instruction tuning trains the model to be able to attribute the generated responses to supporting documents it has seen during pretraining. The pretraining stage will involve all documents by nature, but the instruction tuning step is restricted to a subset of the pretraining documents. This restriction is due to the potential cost of curating instruction tuning data from all the pretraining documents, that is in addition to the training overhead incurred by instruction tuning. After training, we measure out-of-domain (OOD) attribution: whether the model can attribute knowledge to documents that are only included in the continual pretraining data but not in the instruction tuning data. We therefore split the pretraining corpus into in-domain and OOD subsets. The in-domain data is used to create attribution training examples, while the OOD documents are used for evaluation.
The continual pretraining phase has two goals: (i) memorizing knowledge via next-word prediction (same as established LLM pretraining), and (ii) associating knowledge within a source document with its ID to enable OOD attribution. We aim to achieve the second goal by injecting the document ID into the document before training. An important consideration is the location and frequency of injecting the document ID. Formally, given a pretraining corpus of documents {x (1) , x (2) , ..., x (m)} and their corresponding IDs {c (1) , c (2) , ..., c (m)} where each x (i) is a sequence of tokens x (i) = (x (i) 1 , x (i) 2 , ..., x (i) |xi |), and each c (i) = (c (i) 1 , c (i) 2 , ..., c (i) |ci |) is a sequence of tokens of its identifier. Our pretraining aims to learn the language model parameters θ that maximize the objective maxθ ∑ m i=1 log P(xˆ (i) ; θ) , where xˆ (i) is the ID-injected version of the document x (i) . We inject the doc ID into document a x with different strategies, each of which corresponds to a different xˆ. 2 To maximize GPU utilization during continual pretraining, the typical practice packs several pretraining documents within a single training sequence separated by the end-of-sentence token. As a result, the doc ID tokens for a certain document will naturally attend to preceding tokens from other documents. The setup in the previous section train the model to recall the doc ID immediately after the answer.

Figure 6: An example of **research context construction**.

Research Context Annotation — Main Experiment Setup and Results
Paper Title: Source-Aware Training Enables Knowledge Attribution in Language Models
url: https://arxiv.org/pdf/2404.01019v2

Main Experiment Setup and Results: To have a controlled experimental setting, we rely on pretraining knowledge in the form of atomic synthetic facts. We now describe how we construct BIOCITE—a synthetic pretraining corpus. BIOCITE is based on the BioS dataset (Zhu & Li, 2023), which is a collection of biographies of fake people where each biography lists six different facts about each person: birthdate, birth city, study major, university, employer, and work city. Each attribute is described using a corresponding template. For example, the birth city fact is described by " was born in ." To avoid co-reference issues when sampling facts, the person's full name is mentioned in all the facts. To simulate realistic pretraining data that often include facts about different entities, we construct each document in BIOCITE as a collection of facts from at least two different biographies in BioS. More particularly, to construct one document d, we first sample the number of biographies nd ~ Uniform([2, · · · , NMaxNBio]). Then, we sample nd biographies from BioS without replacement. Finally, we sample a random number of facts from each one in the nd biographies and combine these to form the document. We allow the same combination of biographies to create a document only once and allow each fact to appear only once in BIOCITE. In our experiments, we generate 100K documents in total using NMaxNBio = 4.
The pretraining corpus is split 50-50 into in-domain and OOD subsets, respectively. Training is done over 80% of the in-domain question, and we show performance in the remaining 20K. OOD evaluation is performed over 20K questions randomly sampled from the OOD documents. The QA performance is evaluated using the token exact match (EM) with the gold answer. During inference, we prompt the model and let it generate a response first, then append the special token and continue decoding until the model generates the token. We use constrained beam search Cao et al. (2021); Tay et al. (2022) to force the model to generate doc IDs that appeared in the pretraining data. We evaluate attribution by measuring whether the cited document supports the questionanswer pair. Precisely, we measure the gold document ID recall over cases where the answer is correct, where recall is evaluated using Hits@k with k ∈ {1, 10}, which measures whether the gold ID is in the top k beams. To monitor the impact of our attribution training on the model quality, we monitor the perplexity over Wikitext-v2 (Merity et al., 2017) during training (, as done in previous work (Radford et al., 2019), **prior work, removed**). The model we use for all experiments is TinyLLama 1.1B (Zhang et al., 2024),7 which we pretrain for 10 epochs with a learning rate of 8 × 10−5 and instruction-tuning for 3 epochs with a learning rate of 1 × 10−5 . During both pretraining and fine-tuning, we apply a linear decay scheduler and use a batch size of 128, a weight decay of 0.02, and a learning rate warm-up of one epoch.
We start by evaluating the QA performance on OOD quuestions and the answer match over BIOCITE with different document ID injection strategies. The model can achieve OOD answer match > 80%, showing that the model has well memorized the pretraining knowledge. We also note that DOC-BEGIN achieves much worse QA performance than other strategies, and we hypothesize that DOCBEGIN conditions the model to expect the ID when citing knowledge, causing a mismatch during inference when the ID is absent. The ID injection strategy plays a major role in OOD attribution achieved by source-aware training. Placing the ID only once with DOC-BEGIN or DOC-END performs poorly. (We hypothesize that both cases train the model … **analysis, removed**). (Chain-of-thought attribution helps, **ablation study, removed**) REPEAT may be unfavorable, since the number of pretraining tokens will noticeably increase by about 80 tokens, bringing additional training overhead. Besides, the model quality will be negatively impacted since document IDs are not natural text, which is reflected in the perplexity over Wikitext-v2. (The question here is whether source-aware training can yield OOD attribution …, **ablation study, removed**) The results above suggest that source-aware training can teach the model to attribute its parametric knowledge to their pretraining sources, with one key choice to consider: the doc ID injection strategy. (Another key component is document augmentation, …, **ablation study, removed**)

Annotated Main Experiment Setup and Results:
To have a controlled experimental setting, we rely on pretraining knowledge in the form of atomic synthetic facts. We now describe how we construct BIOCITE—a synthetic pretraining corpus. BIOCITE is based on the BioS dataset, which is a collection of biographies of fake people where each biography lists six different facts about each person: birthdate, birth city, study major, university, employer, and work city. Each attribute is described using a corresponding template. For example, the birth city fact is described by " was born in ." To avoid co-reference issues when sampling facts, the person's full name is mentioned in all the facts. To simulate realistic pretraining data that often include facts about different entities, we construct each document in BIOCITE as a collection of facts from at least two different biographies in BioS. More particularly, to construct one document d, we first sample the number of biographies nd ~ Uniform([2, · · · , NMaxNBio]). Then, we sample nd biographies from BioS without replacement. Finally, we sample a random number of facts from each one in the nd biographies and combine these to form the document. We allow the same combination of biographies to create a document only once and allow each fact to appear only once in BIOCITE. In our experiments, we generate 100K documents in total using NMaxNBio = 4.
The pretraining corpus is split 50-50 into in-domain and OOD subsets, respectively. Training is done over 80% of the in-domain question, and we show performance in the remaining 20K. OOD evaluation is performed over 20K questions randomly sampled from the OOD documents. The QA performance is evaluated using the token exact match (EM) with the gold answer. During inference, we prompt the model and let it generate a response first, then append the special token and continue decoding until the model generates the token. We use constrained beam search to force the model to generate doc IDs that appeared in the pretraining data. We evaluate attribution by measuring whether the cited document supports the questionanswer pair. Precisely, we measure the gold document ID recall over cases where the answer is correct, where recall is evaluated using Hits@k with k ∈ {1, 10}, which measures whether the gold ID is in the top k beams. To monitor the impact of our attribution training on the model quality, we monitor the perplexity over Wikitext-v2 during training. The model we use for all experiments is TinyLLama 1.1B,7 which we pretrain for 10 epochs with a learning rate of 8 × 10−5 and instruction-tuning for 3 epochs with a learning rate of 1 × 10−5 . During both pretraining and fine-tuning, we apply a linear decay scheduler and use a batch size of 128, a weight decay of 0.02, and a learning rate warm-up of one epoch.
We start by evaluating the QA performance on OOD quuestions and the answer match over BIOCITE with different document ID injection strategies. The model can achieve OOD answer match > 80%, showing that the model has well memorized the pretraining knowledge. We also note that DOC-BEGIN achieves much worse QA performance than other strategies, and we hypothesize that DOCBEGIN conditions the model to expect the ID when citing knowledge, causing a mismatch during inference when the ID is absent. The ID injection strategy plays a major role in OOD attribution achieved by source-aware training. Placing the ID only once with DOC-BEGIN or DOC-END performs poorly. REPEAT may be unfavorable, since the number of pretraining tokens will noticeably increase by about 80 tokens, bringing additional training overhead. Besides, the model quality will be negatively impacted since document IDs are not natural text, which is reflected in the perplexity over Wikitext-v2. The results above suggest that source-aware training can teach the model to attribute its parametric knowledge to their pretraining sources, with one key choice to consider: the doc ID injection strategy.

Figure 7: An example of **research context construction**.

Ablation Study Annotation
Paper Title: Source-Aware Training Enables Knowledge Attribution in Language Models
url: https://arxiv.org/pdf/2404.01019v2

Ablation Study: We compare two types of data augmentation methods: document and fact augmentation, and the goal is to assess which type of augmentation is necessary for OOD attribution. Document augmentation is done by permuting the facts within a document Naug times and is what our experiment so far have relied on. Fact augmentation duplicates the facts in a document in Naug different random documents. There are two pretraining settings, where Naug is varied and where Naug = 1 meaning no augmentation. In large-scale pretraining setups, this is achieved by scaling the pretraining data but as we study attribution on a smaller scale, we achieve the same effect of redundancy via data augmentation. We start by evaluating the QA performance on OOD quuestions over BIOCITE. While answer match improves using fact-level augmentation, Hits@1 remains the same and only improves when we apply document augmentation. Document augmentation appears necessary for the model to associate the doc ID with the facts in the document.

Annotation:
Research Objective: The goal for this ablation study is to assess which type of augmentation is necessary for OOD attribution.
Experiment Process: We compare two types of data augmentation methods: document and fact augmentation. Document augmentation is done by permuting the facts within a document Naug times and is what our experiment so far have relied on. Fact augmentation duplicates the facts in a document in Naug different random documents. We incorporate two types of data augmentation methods in large-scale pretraining setups by scaling the pretraining data. We evaluate the QA performance on OOD questions over BIOCITE dataset.
Result Discussion: While answer match improves using fact-level augmentation, Hits@1 remains the same and only improves when we apply document augmentation. Document augmentation appears necessary for the model to associate the doc ID with the facts in the document.

Figure 8: An example of **reference ablation study construction**.

Ablation Study Annotation
Paper Title: Self-Selected Attention Span for Accelerating Large Language Model Inference
url: https://arxiv.org/pdf/2404.09336v1

Ablation Study: Achieved sparsity of self-selected attention Since we allow the LLM to dynamically select important tokens, the attention sparsity dynamically vary during autoregressive inference. In this section, we inspect the attention sparsity for each token during autoregressive generation for the arithmetic evaluation task. We analyze evaluation data with ground truth output length between 768 and 1024. We select the first three runs and record the achieved attention sparsity for each output token. The attention sparsity is calculated as the number of ignored tokens divided by the total number of already generated tokens, including the prompt tokens. We plot attention sparsity as a function of the position of the generated token during inference. We observe considerable attention sparsity during autoregressive generation. Specifically, the average attention sparsity over all token positions and across three runs is 47.3%. The attention sparsity generally increases with growing token positions, because the larger number of previously generated tokens, the more opportunities to selectively ignore irrelevant tokens. Occasional dips to 0% occur when generating references to anchors, which require full-context attention. Conclusions. We observe considerable attention sparsity on average and find that LLMs dynamically adjust the attention sparsity during autoregressive generation.

Annotation:
Research Objective: In this section, we inspect the attention sparsity for each token during autoregressive generation for the arithmetic evaluation task. Method.
Experiment Process: Since we allow the LLM to dynamically select important tokens, the attention sparsity dynamically vary during autoregressive inference. We analyze evaluation data with ground truth output length between 768 and 1024. We select the first three runs and record the achieved attention sparsity for each output token. The attention sparsity is calculated as the number of ignored tokens divided by the total number of already generated tokens, including the prompt tokens. We plot attention sparsity as a function of the position of the generated token during inference.
Result Discussion: We observe considerable attention sparsity during autoregressive generation. Specifically, the average attention sparsity over all token positions and across three runs is 47.3%. The attention sparsity generally increases with growing token positions, because the larger number of previously generated tokens, the more opportunities to selectively ignore irrelevant tokens. Occasional dips to 0% occur when generating references to anchors, which require full-context attention. Conclusions. We observe considerable attention sparsity on average and find that LLMs dynamically adjust the attention sparsity during autoregressive generation.

Figure 9: An example of **reference ablation study construction**.

Ablation Study Annotation
Paper Title: PetKaz at SemEval-2024 Task 3: Advancing Emotion Classification with an LLM for Emotion-Cause Pair Extraction in Conversations
url: https://arxiv.org/pdf/2404.05502v1

Ablation Study: For this stage, we fine-tune GPT-3.5. As a system's input, we provide the prompt consisting of an instruction, Ut−1, and Ut. This particular prompt was selected during the preliminary prompt engineering stage. The assistant's output consists of one word – the emotion type. We also note that preliminary experiments showed that the LLM performed poorly in zeroshot and few-shot settings on the emotion detection task, at least on the ECF dataset. Therefore, we had to fine-tune it. Table 2 overviews the performance of emotion classification using GPT-3.5 across different paradigms: zero-shot, few-shot, and fine-tuning. We note that zero- and few-shot settings use the same prompt, with the few-shot setting including one handpicked example per each emotion type. As expected, finetuning yields the best results on all emotion types and overall. Interestingly, few-shot prompting performs worse than zero-shot, which suggests that examples hamper the model's understanding of emotion types instead of improving it.

Annotation:
Research Objective: We examine three configurations on LLM training settings for emotion classification to determine the significance of fine-tuning.
Experiment Process: We utilize three different paradigms, zero-shot, few-shot, and fine-tuning, to train GPT-3.5 on the task of emotion classification, and analyze its performance. As a system's input, we provide the prompt consisting of an instruction, Ut−1, and Ut. We note that zero- and few-shot settings use the same prompt, with the few-shot setting including one handpicked example per each emotion type. This particular prompt was selected during the preliminary prompt engineering stage. The assistant's output consists of one word – the emotion type. Experiments are conducted on the ECF dataset.
Result Discussion: Finetuning yields the best results on all emotion types and overall. Preliminary experiments showed that the LLM performed poorly in zeroshot and few-shot settings on the emotion detection task. Interestingly, few-shot prompting performs worse than zero-shot, which suggests that examples hamper the model's understanding of emotion types instead of improving it.

[Table 2] <table border="1" cellpadding="5" cellspacing="0"><tr><th>Approach</th><th>neutral</th><th>anger</th><th>disgust</th><th>fear</th><th>joy</th><th>sadness</th><th>surprise</th><th>macro</th><th>w-avg</th></tr><tr><td>Zero-shot</td><td>0.61</td><td>0.43</td><td>0.30</td><td>0.32</td><td>0.54</td><td>0.47</td><td>0.50</td><td>0.45</td><td>0.54</td></tr><tr><td>Few-shot</td><td>0.57</td><td>0.49</td><td>0.31</td><td>0.34</td><td>0.54</td><td>0.37</td><td>0.41</td><td>0.43</td><td>0.51</td></tr><tr><td>Fine-tuning</td><td>0.70</td><td>0.57</td><td>0.42</td><td>0.51</td><td>0.63</td><td>0.52</td><td>0.66</td><td>0.57</td><td>0.64</td></tr></table> Table 2: F₁ scores on emotion classification with GPT-3.5 across different approaches.

Figure 10: An example of **reference ablation study construction**.

Error Analysis: Misalignment with Research Context
Paper Title: MoDE: CLIP Data Experts via Clustering
url: https://arxiv.org/pdf/2404.16030v1

Ablation Study Design generated by GPT-4o about Data Experts
To understand the contribution of Data Experts in the MoDE framework, we design an ablation study where we systematically remove data experts ❌ and compare it with the full MoDE model. Specifically, we will test ❌ MoDE with zero, two, and four data experts. By evaluating performance on tasks like zero-shot image classification and retrieval, we will quantify the impact of having fewer specialized experts. This step-by-step reduction helps isolate the specific benefit each expert provides and highlights the importance of separating training data into clusters to handle false negatives effectively.

Relevant text in "Research Context": We set the number of fine-grained clusters, and report performance for both MoDE-2 and MoDE-4 below to directly show the improvement by increase the number of data expert models ✓ on all evaluation tasks.

Error analysis by human experts: The main experiment did not define a "full MoDE model.", resulting in misalignment with Research Context. These data experts were continuously trained, updated, and added, and could not be reduced to zero, resulting in misinformation.

Figure 11: An error example of **Misalignment with Research Context**.

Error Analysis: Ambiguity and Difficulty in Reproduction
Paper Title: AISPACE at SemEval-2024 task 8: A Class-balanced Soft-voting System for Detecting Multi-generator Machine-generated Text
url: https://arxiv.org/pdf/2404.00950v1

Ablation Study about Loss Function Analysis
This ablation study aims to evaluate the impact of different loss functions on the effectiveness of our model in detecting machine-generated text. We will compare the standard Cross Entropy loss with our proposed class-balanced weighted loss function. Each model will be trained separately using both loss functions on the same dataset. ❌ We will measure performance based on accuracy and other relevant metrics. ❌ Through this analysis, we aim to understand how the class imbalance affects model predictions and whether our proposed loss function significantly improves detection across different classes. This insight will help us refine our approach to loss function selection in future work.

Relevant text in "Research Context": For the baseline experiments on Roberta-large, we utilized the official subtaskB dataset and the merged data ✓ for separate training to determine the effectiveness of the merged data. For fine-tuning other models, we only used the merged data, which has been proven effective in the baseline experiment. The original subtaskB training data consists of 71027 items, while merging data results in a total of 127,755 items. We divided the data into train and val sets in an 8:2 ratio, and the original subtaskB dev set was kept as the dev set, which contains 3,000 items. No additional data was used for either training or evaluation. The official evaluation metric for the SubtaskB is accuracy. The experiments result in this paper is based on accuracy of dev set. ✓

Error analysis by human experts: The generation does not incorporate relevant datasets or tools necessary for replication from the "Research Context", resulting in ambiguity and difficulty in reproduction.

Figure 12: An error example of **Ambiguity and Difficulty in Reproduction**.

Error Analysis: Partial Ablation or Incomplete Experimentation
Paper Title: Global Contrastive Training for Multimodal Electronic Health Records with Language Supervision
url: http://arxiv.org/pdf/2404.06723v1

Ablation Study about variable-specific encoder and time embedding
To evaluate the significance of the variable-specific encoder and Time2Vec embeddings, we designed an ablation study removing these components. ❌ By doing so, the modified model will rely solely on shared encoder structures and standard positional encoding. We train both the full and the ablated models ❌ under the same conditions and compare their performance on the multitask classification of 9 postoperative complications. The difference in the outcomes will quantify the contribution of these specialized encodings to the model's predictive accuracy, providing insights into their importance in handling high-dimensional, sparse, and irregular EHR time series.

"Reference Ablation Study" in paper document: For encoding each variable separately, ✓ as described in Subsection 3.1.1, we introduced ... The results showed the effectiveness of using a variable-specific encoder indicated by the performance lift, and the model performance of using different types of encoders were similar (For 1-D CNN, transformer, and linear, the mean AUROC score was 0.796, 0.800, and 0.798, respectively)... For examining the effectiveness of time embedding, ✓ experimental results of adding additional time embedding further improved the AUROC by 0.03.

Error analysis by human experts: The instructions specified two components, therefore at least two experimental groups are necessary to reveal the significance of each ablation component. The ablation study generation however, proposes remove both modules at once and compare to the baseline, which results in incomplete ablation design.

Figure 13: An error example of **Partial Ablation or Incomplete Experimentation**.

Error Analysis: Insignificant Ablation Module
Paper Title: AISPACE at SemEval-2024 task 8: A Class-balanced Soft-voting System for Detecting Multi-generator Machine-generated Text
url: https://arxiv.org/pdf/2404.00950v1

Ablation Study about Input Size
We conducted an ablation study to evaluate the impact of input size ✗ on the performance of Longformer, given the dataset's token length distribution. We fine-tuned two versions of the model using input sizes of 512 and 1024 tokens, respectively. By comparing the performance metrics, we aimed to determine the optimal input size for detecting machine-generated text. This approach helped us assess how the length of input sequences affects the model's ability to generalize and effectively identify machine-generated content from different text generators and domains.

Relevant text in "Research Context": Furthermore, after the merging of data, we analysed the token length of the dataset. The majority of the token length in the training text falls within the range of 0- 1000, whereas the length of the development text is mostly between 0-500. So our system tested input size of 512 and 1024 tokens ✓ in Longformer model.

Error analysis by human experts: The generation is focused on evaluating the impact of input size. However, input size is only a trivial factor in the model design in the research context. It's value is determined by the majority of the development text length. Therefore this ablation design focused on a trivial model parameter is insignificant.

Figure 14: An error example of **Insignificant Ablation Module**.

Error Analysis: Inherent Logical Inconsistencies
Paper Title: Global Contrastive Training for Multimodal Electronic Health Records with Language Supervision
url: http://arxiv.org/pdf/2404.06723v1

Ablation Study about Multimodal Contrastive Loss
In this ablation study, we aim to evaluate the contribution of the global contrastive loss ✓ to the overall performance of the multimodal EHR modeling framework. Specifically, we will remove the global contrastive loss component from the model and observe the impact on the model's ability to align multimodal feature representations with discharge summaries. This study will help us understand the significance of the contrastive loss ✗ in enhancing the model's learning of discriminative multimodal features and its role in improving the model's effectiveness in capturing the interconnected relationships between medical time series and clinical notes.

Error analysis by human experts: The generation initially focused on the multimodal (global) contrastive loss as instructed. However, the conclusion shifted to discussing the significance of contrastive loss in general. This ablation design contains inherent logical inconsistencies.

Figure 15: An error example of **Inherent Logical Inconsistencies**.

User Study Example
Paper Title: Self-Selected Attention Span for Accelerating Large Language Model Inference
url: https://arxiv.org/pdf/2404.09336v1

Initial Generation by GPT-4o
Ablation Study about Attention Mechanisms
The role of the attention mask is to determine data dependency between tokens. By removing or disabling the attention mask module, we can evaluate the extent to which it influences the efficiency of LLMs in sparse attention by directly comparing model performance and computational cost, particularly focusing on self-attention computation time and memory access patterns without the pre-filtering capability of the mask. This study can help clarify the necessity and impact of dynamic attention masking in speeding up inference in various task settings.

Human Evaluation Scores:
Faithfulness: 2, Attention mask is not directly stated in the research context, which uses attention mechanisms instead. The logical relation between attention mask and attention mechanisms not explained, poor alignment.
Soundness: 2, logically incoherent, for example, how to remove or disable the attention mask module?
Importance: 4, lack comprehensive experimental aspects or insights.

User Feedback:
This ablation study generation proposes removing the attention mask module. However, exactly how to remove this module, will the model still work with the attention mask module removed, can this ablation study reveal the significance of the attention mechanism? Is there another way to design an ablation study about the attention mechanism?

Revised Generation by GPT-4o
Ablation Study about Attention Mechanisms
We propose an ablation study to assess the impact of the attention mask on model efficiency. Instead of completely removing the attention mask, we will vary its settings to examine how different levels of masking affect computational performance and model accuracy. Specifically, we will compare three scenarios: no masking, static masking (where the mask is fixed), and dynamic masking (where the mask is adaptively adjusted based on context). This approach allows us to evaluate the role of the attention mask in optimizing self-attention computation, focusing on how masking strategies influence memory access patterns and overall inference speed. We will use benchmark datasets and compare results against the baseline to ensure the reliability of findings.

Human Evaluation Scores:
Faithfulness: 5
Soundness: 4, The new generation improves upon the initial generation, adding notes about how to realize the three scenarios, such as static masking and dynamic masking.
Importance: 5

Figure 16: An example where user feedback enhances ablation study generation quality.

User Study Example
Paper Title: MoDE: CLIP Data Experts via Clustering
url: https://arxiv.org/pdf/2404.16030v1

Initial Generation:
Ablation Study about Clustering Approach
In this ablation study, we will disable the two-step K-means clustering and use a single clustering step to group the training data. This simplification will help us understand the significance of fine-grained versus coarse-grained clustering in enhancing the model's ability to handle false negatives and create challenging negative examples. By evaluating the performance ✗ of the MoDE framework with a single clustering approach, we aim to demonstrate the necessity of our hierarchical clustering strategy ✗ for better downstream task performance and overall training effectiveness.

— gpt-4o

Human Evaluation Scores:
Faithfulness: 2, the research context doesn't explicitly states the two-step clustering method as a hierarchical strategy, the narration doesn't align well with the research context.
Soundness: 2, logically incomplete, the experiment process is inconsistent, for example, how to evaluate model performance after ablation.
Importance: 3, the experiment process is not sufficient to derive the conclusion..

User Feedback:
This ablation study generation proposes simplifying the clustering strategy to reveal the significance of the two-step approach. However, exactly how to design and implement the experiment, which dataset and evaluation metrics should we use, what is the specific contribution of the clustering strategy instead of its significance in general? Is there another way to design an ablation study about the clustering strategy?

Revised Generation:
Ablation Study about Attention Strategy
We propose an ablation study to evaluate the impact of the two-step clustering strategy ✓ in the MoDE framework. The study involves comparing the standard two-step clustering approach (fine-grained followed by coarse-grained) against a single-step clustering approach. We will use the MetaCLIP datasets (400M and 2.5B image-caption pairs) for training and standard benchmarks (ImageNet, COCO, Flickr30k) for evaluation. Metrics ✓ will include zero-shot classification accuracy, retrieval performance (Recall@K), and training efficiency (GPU hours). Additionally, we will assess cluster coherence by analyzing intra-cluster and inter-cluster distances. This study aims to clarify the contribution of hierarchical clustering in reducing false negatives and improving task-specific performance.

— gpt-4o

Human Evaluation Scores:
Faithfulness: 5
Soundness: 5
Importance: 5

Figure 17: An example where user feedback enhances ablation study generation quality.

C Meta-Evaluation Experiments

To evaluate the performance of automatic metrics, the human evaluation result on the same evaluation target is considered the gold standard, and metric performance is measured by the correlation between the human evaluation scores and automatic metric scores. For ABGEN-EVAL, such correlations are calculated at the *system* level and *instance* level. Specifically, given n input scientific papers and m ablation study generation systems, the human evaluation and an automatic metric result in two n-row, m-column score matrices H, M respectively. The *system*-level correlation is calculated on the aggregated system scores:

$$r_{\text{sys}}(H, M) = \mathcal{C}(\bar{H}, \bar{M}), \qquad (2)$$

where $\bar{H}$ and $\bar{M}$ contain m entries which are the average system scores across n data samples (e.g., $\bar{H}_0 = \sum_i H_{i,0}/n$), and $\mathcal{C}$ is a function calculating a correlation coefficient (e.g., the Pearson's correlation coefficient). In contrast, the *instance*-level correlation is an average of sample-wise correlations:

$$r_{\text{sum}}(H, M) = \frac{\sum_i \mathcal{C}(H_i, M_i)}{n}, \qquad (3)$$

where H_i, M_i are the evaluation results on the i-th data sample.

Automated Evaluation Systems	Faith.	Sound.	Import.	Avg.
Traditional Metrics				
BLEU	-0.014	0.092	0.063	0.047
ROUGE-L	0.087	-0.002	0.045	0.043
BERTScore	0.028	0.071	0.039	0.046
LLM Pointwise Scoring				
GPT-4o-mini (Criteria-specific)				
Coarse instruct + CoT	-0.02	0.12	0.17	0.09
Ref. + Fine-grained instruct	-0.04	0.18	0.08	0.07
Coarse instruct	0.03	0.13	0.05	0.07
Fine-grained instruct	0.06	0.03	0.09	0.06
Ref. + Coarse instruct + CoT	0.08	0.03	0.06	0.06
Ref. + Coarse instruct	0.03	0.07	0.04	0.05
Ref. + Fine-grained instruct + CoT	-0.01	0.08	0.00	0.03
Fine-grained instruct + CoT	-0.07	0.04	0.07	0.01
GPT-4o-mini (Overall)				
Ref. + Fine-grained instruct + CoT	0.008	0.046	0.010	0.021
GPT-4o (Criteria-specific)				
Fine-grained instruct + CoT	0.134	0.158	0.135	0.176
Fine-grained instruct	0.108	0.146	0.110	0.121
Ref. + Fine-grained instruct + CoT	0.158	0.116	-0.002	0.124

Table 10: Pearson correlations at the instance level between automated evaluation and human assessments.

LLM Pointwise Scoring with CoT Reasoning

[System Input]
Evaluate the ablation study design based on the given instructions.

Here are some rules of the evaluation:
{rules}

[User Input]
Research Context:
{research_context}

Reference Ablation Study Design:
{reference}

Ablation Study Design about {module_name} (To Evaluate)
{model_output}

You should provide a brief explanation of your evaluation, an then assign a score (an integer) on a scale of 1 to 5, where a higher score signifies better {criteria}. Your response should be in the format of a JSON object, as follows: {"reason": <a brief explanation>, "score": <an integer between 1 and 5>}.

Your Response:

Figure 18: Prompt for LLM Pointwise Scoring with CoT Reasoning (reference optional).

LLM Pointwise Scoring without CoT Reasoning

[System Input]
Evaluate the ablation study design based on the given instructions.

Here are some rules of the evaluation:
{rules}

[User Input]
Research Context:
{research_context}

Reference Ablation Study Design:
{reference}

Ablation Study Design about {module_name} (To Evaluate)
{model_output}

You should only assign a score (an integer) on a scale of 1 to 5, where a higher score signifies better {criteria}. Your response should be in the format of a JSON object, as follows: {"score": <an integer between 1 and 5>}.

Your Response:

Figure 19: Prompt for LLM Pointwise Scoring without CoT Reasoning (The feature of "reference" is optional).

LLM Pairwise Comparison with CoT Reasoning

[System Input]
Evaluate the ablation study designs based on the given instructions.

Here are some rules of the evaluation:
{rules}

[User Input]
Research Context:
{research_context}

Reference Ablation Study Design:
{reference}

Ablation Study Design (A) about {module_name} (To Evaluate)
{model_output_1}

Ablation Study Design (B) about {module_name} (To Evaluate)
{model_output_2}

Before selecting the Output (A) or Output (B) that signifies better {criteria}, give a brief explanation for your selection. Your response should be in the format of a JSON object, as follows: {"reason": <a brief explanation>, "decision": <only "A" or "B" or "Tie">}.

Your Response:

Figure 20: Prompt for LLM Pairwise Comparison with CoT Reasoning (The feature of “reference” is optional).

Faithfulness:
1. Verify that the generated ablation study correctly aligns with the focused module and addresses the ablation module. (5 points)
2. Ensure the generated ablation study maintains consistency with the research paper, particularly avoiding conflicts with Baseline Experiment Settings. (5 points)
3. Detect any discrepancies between the generated ablation study and the Experiment Settings in the research paper. (3 points)
4. Identify significant misalignment or misinformation in the generated ablation study based on the focused module or research paper. (1 point)
5. Penalize from the labeled score if the criterion is unmet; for example, a failure in a 5-point criterion should result in no more than 4 points.
6. Assign scores so that the generated ablation study meets or exceeds the performance of all criteria below the given score. For example, a 4-point score should perform no worse than a 3-point criterion.
7. Prefer concise answers; deduct points if the generated ablation study exceeds 150 words, ensure that the final score does not exceed 4.

Soundness:
1. Evaluate if the generated ablation study presents a clear, specific research objective. (5 points)
2. Check if the generated ablation study lacks a clear objective, resulting in a confusing or wordy narrative. (1 point)
3. Verify that the generated ablation study is logically coherent and practically feasible to implement. (5 points)
4. Assess any gaps in the experiment design of the generated ablation study. (3 points)
5. Penalize from the labeled score if the criterion is unmet; for example, a failure in a 5-point criterion should result in no more than 4 points.
6. Assign scores so that the generated ablation study meets or exceeds the performance of all criteria below the given score. For example, a 4-point score should perform no worse than a 3-point criterion.
7. Prefer concise answers; deduct points if the generated ablation study exceeds 150 words, ensure that the final score does not exceed 4.

Importance:
1. Ensure the generated ablation study addresses a key component of the Baseline model in the research paper. (5 points)
2. Verify that the generated ablation study thoroughly analyzes the module's contribution. (5 points)
3. Confirm if the generated ablation study fails to offer valuable insights for improving model efficiency. (3 points)
4. Assess whether the generated ablation study focuses on an insignificant module. (1 point)
5. Evaluate if the experiment process in the generated ablation study fails to assess the importance of the ablation module. (1 point)
6. Penalize from the labeled score if the criterion is unmet; for instance, a 5-point criterion failure should result in no more than 4 points.
7. Assign scores to the generated ablation study such that it outperforms all criteria below the given score. For example, an answer that receives 4 points should perform no worse than the criteria labeled (3 points).
8. Prefer concise answers; deduct points if the generated ablation study exceeds 150 words, ensure that the final score does not exceed 4.

Overall:
1. Verify if the generated ablation study correctly answers the focused module or task (5 points).
2. Assess the overall feasibility of the generated ablation study referencing research paper (5 points).
3. Ensure that the generated ablation study provides valuable insights for improving model efficiency in the research paper (5 points).
4. Check if the generated ablation study contain significant defects in addressing focused module, maintaining consistent narrative or providing insights (1 points).
5. Scores between 1 and 5, should be interpreted as intermediate values between the defined criteria.
6. Prefer concise answers; deduct points if the generated ablation study exceeds 150 words, ensure that the final score does not exceed 4.

Figure 21: Fine-grained evaluation instructions for the 4 criteria in LLM pointwise scoring.

Faithfulness:
A good ablation study should align with the context of the paper, avoiding the introduction of misinformation or contradictions to the baseline experiments presented in the original document.

Soundness:
Generations should be feasible, logically consistent, and maintain a consistent topic and narrative throughout. A good ablation study should adhere to the definition of ablation studies, focusing on validating the significance of the given model component.

Importance:
Evaluate the significance of the generated ablation study design, considering the relevance of the specific module to the model design and assessing the coverage of available datasets, metrics, or experimental groups within the paper's context.

Overall:
A good ablation study should feasibly address the focused module and provide valuable insights for improving model efficiency in the research paper.

Figure 22: Coarse evaluation instructions for the 4 criteria in LLM pointwise scoring.

Faithfulness:
1. Ensure that the generated ablation study correctly aligns with the focused module and properly addresses the ablation module.
2. Confirm that the generated ablation study remains consistent with the research paper, especially avoiding any conflicts with the Baseline Experiment Settings.
3. Identify any discrepancies between the generated ablation study and the Experiment Settings outlined in the research paper.
4. Detect any significant misalignment or misinformation in the generated ablation study based on the focused module or research paper.
5. Prefer the generated ablation study that does not exceed 150 words, while still providing a concise and clear answer.

Soundness:
1. Evaluate whether the generated ablation study presents a clear and specific research objective.
2. Determine if the ablation study lacks a clear objective, leading to a confusing or overly wordy narrative.
3. Verify that the ablation study is logically coherent and practically feasible to implement.
4. Assess any gaps in the experimental design of the generated ablation study.
5. Prefer the generated ablation study that does not exceed 150 words, while still providing a concise and clear answer.

Importance:
1. Ensure the generated ablation study focuses on a key component of the Baseline model as described in the research paper.
2. Verify that the ablation study thoroughly analyzes the contribution of the module being ablated.
3. Insight Evaluation: Confirm if the ablation study fails to provide valuable insights for improving model efficiency.
4. Significance Check: Assess whether the ablation study targets an insignificant module within the Baseline model.
5. Experiment Assessment: Evaluate if the experimental process in the ablation study fails to assess the importance of the ablated module.
6. Prefer the generated ablation study that does not exceed 150 words, while still providing a concise and clear answer.

Overall:
1. Verify if the generated ablation study correctly answers the focused module.
2. Assess the overall feasibility of the generated ablation study referencing research paper.
3. Ensure that the generated ablation study provides valuable insights for improving model efficiency in the research paper.
4. Check if the generated ablation study contain significant defects in addressing focused module, maintaining consistent narrative or providing insights.
5. Prefer the generated ablation study that does not exceed 150 words, while still providing a concise and clear answer.

Figure 23: Fine-grained evaluation instructions for the 4 criteria in LLM pairwise comparison.